

2024 Ph H2 Q9

Section: Particles and Waves

Topic: Wave-Particle Duality (Photoelectric Effect)

Summary of Question: UV light (frequency 1.25×10^{15} Hz) shines on a zinc plate in a vacuum tube with a mesh anode.

(a) Calculate photon energy; (a)(ii) explain why a current is produced; (b) state and justify effect on current when lamp is moved closer; (c) state and justify effect on current when supply polarity is reversed.

(a)(i) Photon energy

Relationship: $E = hf$, where $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$

Substitution: $E = (6.63 \times 10^{-34})(1.25 \times 10^{15})$

Calculation: $E = 8.29 \times 10^{-19} \text{ J} \approx 8.3 \times 10^{-19} \text{ J}$

Final Answer: $E = 8.3 \times 10^{-19} \text{ J}$

(a)(ii) Why a current is produced

Photon energy ($8.29 \times 10^{-19} \text{ J}$) is greater than zinc's work function ($5.81 \times 10^{-19} \text{ J}$), so electrons are emitted.

The electrons are attracted to the positive mesh, completing the circuit and producing a measurable current.

Final Answer: $E > \phi$ so electrons are emitted and collected at the mesh, producing a current.

(b) Moving the lamp closer

Conclusion: The current increases.

Justification: A closer lamp means more photons per second strike the zinc, ejecting more electrons per second.

Final Answer: Current increases because more photons strike the zinc plate per second.

(c) Reversing the polarity

Conclusion: The current becomes zero.

Justification: The zinc plate becomes positive, attracting electrons back so they do not reach the mesh.

Final Answer: Current is zero because electrons are attracted back to the positively charged zinc plate.

Revision Tips

- Standard 3-mark calculation: relationship $\rightarrow$ substitution $\rightarrow$ final answer (with units & sig figs).
- Photoemission condition: $hf \geq \phi$ (or $f \geq$ threshold frequency).

- Increasing intensity → more current (not higher electron KE).
Increasing frequency → determines if emission occurs.